

Solving Quadratic Equations by Factoring — Version A

Name: _____

Date: _____

Period: _____

Directions: Solve each equation by factoring. Show all work. Write your solutions as exact values.

1. $(2x - 5)(x + 3) = 0$

Answer: _____

2. $2x^2 + 7x + 3 = 0$

Answer: _____

3. $3x^2 + 2 = 7x$

Answer: _____

4. $12x^2 + 10x = 8$

Answer: _____

Solving Quadratic Equations by Factoring — Version B

Name: _____

Date: _____

Period: _____

Directions: Solve each equation by factoring. Show all work. Write your solutions as exact values.

1. $(x - 4)(5x + 2) = 0$

Answer: _____

2. $3x^2 + 10x + 8 = 0$

Answer: _____

3. $4x^2 - 3 = 11x$

Answer: _____

4. $15x^2 + 6 = 21x$

Answer: _____

Solving Quadratic Equations by Factoring — Version C

Name: _____

Date: _____

Period: _____

Directions: Solve each equation by factoring. Show all work. Write your solutions as exact values.

1. $(4x + 3)(x - 7) = 0$

Answer: _____

2. $5x^2 - 17x + 6 = 0$

Answer: _____

3. $2x^2 + 15 = 13x$

Answer: _____

4. $20x^2 + 10x = 30$

Answer: _____

Answer Key — Version A

1. $x = 5/2, x = -3$

2. $(2x + 1)(x + 3) = 0 \rightarrow x = -1/2, x = -3$

3. $3x^2 - 7x + 2 = 0 \rightarrow (3x - 1)(x - 2) = 0 \rightarrow x = 1/3, x = 2$

4. $2(6x^2 + 5x - 4) = 0 \rightarrow 2(2x - 1)(3x + 4) = 0 \rightarrow x = 1/2, x = -4/3$

Answer Key — Version B

1. $x = 4, x = -2/5$

2. $(3x + 4)(x + 2) = 0 \rightarrow x = -4/3, x = -2$

3. $4x^2 - 11x - 3 = 0 \rightarrow (4x + 1)(x - 3) = 0 \rightarrow x = -1/4, x = 3$

4. $3(5x^2 - 7x + 2) = 0 \rightarrow 3(5x - 2)(x - 1) = 0 \rightarrow x = 2/5, x = 1$

Answer Key — Version C

1. $x = -3/4, x = 7$

2. $(5x - 2)(x - 3) = 0 \rightarrow x = 2/5, x = 3$

3. $2x^2 - 13x + 15 = 0 \rightarrow (2x - 3)(x - 5) = 0 \rightarrow x = 3/2, x = 5$

4. $10(2x^2 + x - 3) = 0 \rightarrow 10(2x + 3)(x - 1) = 0 \rightarrow x = -3/2, x = 1$